

Chapter 32 — Embryology of the Human Genitourinary Tract

IF YOU READ NOTHING ELSE

- **SRY (sex-determining region Y)**, Yp → Sertoli cells → testis; no SRY → ovary (default)
- **Sertoli** → **MIS (Müllerian-inhibiting substance)** → Müllerian regression wk 8–10; **Leydig** → **testosterone** from wk 8
- **Type 2 5 α -reductase** → DHT (dihydrotestosterone); deficiency → **severe hypospadias, rudimentary prostate**
- **Whole urethral epithelium** = **endoderm (urogenital sinus, UGS)**, glans included
- **Hypospadias** ~1:250; commonest meatus **shaft-glans junction**
- **12 o'clock nerve-free zone** (thickest tunica) → **dorsal plication**
- **Cryptorchidism** ~3% newborn males; hydrocele/hernia commoner
- **46,XY streak gonad** → **remove** (gonadoblastoma)
- **Entire vaginal epithelium** = **UGE** (urogenital sinus epithelium; Bulmer)
- **Weigert-Meyer**: upper pole **caudal-medial** → **ectopic**; lower pole **lateral** → **VUR** (vesicoureteral reflux)

DEVELOPMENT OF THE EXTERNAL GENITALIA

- **Timeline**: indifferent to **8–9 wk** → coronal sulcus **11–12 wk** → folds fused **16 wk** → complete **18 wk** → chordee gone **20 wk**
- **First sign of masculinisation**: ↑ **anogenital distance**
- **Germ layers**: ectoderm → skin/prepuce; mesoderm → corpora; **endoderm** → **urethra**
- **Double zipper**: **canalisation** (proliferation, no apoptosis) → **fold fusion** (shaft)
- **Glans urethra**: **direct canalisation** of plate, not fold fusion
- **Glenister**: ectodermal-intrusion theory of glans urethra — outdated
- **Innervation**: somatic **S2–S4 pudendal**; sympathetic **L1–L2**; parasympathetic **S2–S4**
- **Dorsal nerves**: **11 and 1 o'clock**, fan ventrally to spongiosum; **12 o'clock free**
- **Alcock's canal**: pudendal canal carrying pudendal nerve
- **Hypospadias tetrad**: ectopic meatus, **dorsal hooded foreskin**, ventral skin deficiency, ± curvature
- **Hypospadias** = **developmental arrest**, not deformation
- **Aetiology**: mostly unknown; **genetic susceptibility + endocrine disruptors**
- **Operate if**: mild-moderate curvature → **dorsal 12 o'clock plication**
- **Chordee without hypospadias**: penoscrotal webbing
- **Duplicated urethra**: plate splitting; complete vs **Y-duplication**; **Sonic hedgehog**

DEVELOPMENT OF THE FEMALE EXTERNAL GENITALIA

- **Clitoris** = **penis minus urethra**
- **Canalisation** androgen-independent; **fold fusion** androgen-dependent
- **Clitoral nerves**: densest dorsally; **12 o'clock nerve-free**
- **CAH (congenital adrenal hyperplasia)**: autosomal recessive, ↓ cortisol → virilised 46,XX; urethra can be **male-identical**
- **CAH clue**: **impalpable gonads + hyperpigmentation**

DEVELOPMENT OF THE HUMAN PROSTATE

- **Buds**: **10 wk** (9.5–10) from **FOXA1+ UGE**, gutters **lateral to verumontanum**
- **Branching**: from ~**11 wk**; sets adult **zonal anatomy**
- **Verumontanum**: mesoderm (Wolffian + utricles) meets endoderm (UGE)
- **Six stages**: prebud → buds → branching → canalisation → luminal/basal → secretion
- **Canalisation**: proximal → distal
- **Markers**: basal **TP63, KRT5/14**; luminal **KRT8/18**
- **Smooth muscle**: **ventral first**, then lateral/dorsal

- **DHT**: ~**10 $\times$** AR (androgen receptor) affinity vs testosterone; **5 α R2** in mesenchyme
- **No functional AR** → **no prostate**
- **Mesenchyme instructs epithelium**: UGM (urogenital sinus mesenchyme) induces prostate even in bladder epithelium
- **Squamous metaplasia**: needs **ESR1 in stroma + epithelium**; normal fetal (maternal oestrogen)
- **Jost / Price**: animal proof of androgen-dependent prostate
- **Seminal vesicle**: **Wolffian diverticulum**, **10–12 wk**; side ducts months 4–5; nerves 13 wk

DEVELOPMENT OF THE GONADS

- **Germinal ridge** **4 wk**; **PGCs (primordial germ cells)** **yolk sac wk 3** → hindgut mesentery → ridge
- **Primary sex cords** **wk 6**; seminiferous cords from **mesonephric tubules**
- **Testis cords** ~**6 wk**; **ovarian cortex/medulla** ~**8 wk**
- **Gonads identical 7–8 wk**; testis becomes spherical, ovary oblong
- **Efferent ductules**: **5–12 mesonephric tubules**; full continuity **3rd month**
- **Leydig testosterone**: hCG (human chorionic gonadotrophin) early → fetal pituitary later
- **Wolffian** → **epididymis, vas, seminal vesicle** (wk 8–12)
- **Appendix testis** = **Müllerian**; **appendix epididymis** = **Wolffian**; **paradidymis** = **mesonephric**
- **Ovary**: germ-cell nests **10–11 wk**; primary follicles ~**20 wk**; secondary at **term**; antral postnatal
- **Oocytes**: meiosis I in fetus → **arrest to puberty**
- **Homologues**: Cowper ↔ **Bartholin**; Littre ↔ **Skene**; spongiosum ↔ **vestibular bulb**
- **Gartner duct**: Wolffian remnant, anterolateral vaginal wall
- **Descent**: transabdominal **wk 10–15**; inguinoscrotal from **7th month**; migration **3–5 cm**
- **CSL (cranial suspensory ligament)** regression androgen-dependent; gubernacular swelling androgen-independent
- **Inguinoscrotal phase**: androgen via **genitofemoral nerve** → **CGRP** (calcitonin gene-related peptide)
- **INSL3 (insulin-like peptide 3)**: Leydig; knockout → **intra-abdominal cryptorchidism** only
- **Ovary**: descends **3rd month**; gubernaculum → **ovarian + round ligaments**
- **Retained CSL** → **intra-abdominal testis**; ectopic gubernaculum → perineal/femoral/penile
- **Processus not closed** → **hernia/hydrocele**; **not involuted** → **acquired (ascending) testis**
- **Streak gonad**: **45,X / 46,XX / 46,XY**; no MIS, no testosterone → female; **HRT (hormone replacement)**

- **Ovotestis:** ducts follow ipsilateral gonad; female fertility possible, **no spermatogenesis**
- **PMDS (persistent Müllerian duct syndrome):** 46,XY, **MIS receptor defect**; found at exploration for **bilateral abdominal testes**
- **Müllerian MIS sensitivity:** 9–13 wk

MÜLLERIAN STRUCTURES (FEMALE INTERNAL GENITALIA)

- **Müllerian ducts start wk 5; Wolffian duct = obligatory guidewire**
- **Fusion → uterovaginal canal wk 8; septum gone wk 9**
- **Carnegie stage 22** ≈ 54 days, indifferent stage
- **Vaginal canalisation:** starts ~15 wk, done 18–19 wk
- **Markers:** PAX2 = Müllerian; FOXA1 = UGE; switch at vaginal-cervical junction
- **Koff:** mixed, Müllerian-predominant; **Bulmer:** all UGE — supported
- **Hymen:** UGE-derived
- **Transverse vaginal septum:** failed canalisation, ~1:80,000, middle/upper third junction
- **Imperforate hymen = commonest vaginal obstruction**
- **Didelphys:** failed fusion; **unicornuate:** one duct absent; **septum persists** → obstruction/duplication
- **OHVIRA (obstructed hemi-vagina and ipsilateral renal anomalies):** post-menarche cyclic pain, discharge; acute abdomen/**tubo-ovarian abscess**
- **OHVIRA kidney:** usually **dysplastic**, not agenetic; **true agenesis** → absent hemi-trigone
- **Do not assume agenesis:** ectopic ureter → **vaginal leakage** after septum resection

BLADDER: URETERAL DEVELOPMENT

- **Cloaca divided wk 5–6 by urorectal septum** → anorectal canal + UGS
- **Anorectal malformations:** from **cloacal membrane**, not septum

- **Allantois → urachal remnant**
- **Wolffian entry** divides vesicourethral canal (bladder, pelvic urethra) from caudal UGS
- **Common excretory ducts join UGS wk 5–6;** apoptosis frees ureters
- **Trigone:** mostly **bladder smooth muscle**, minor ureteral input
- **Bud too low** → lateral orifice → **VUR**; **bud too high** → caudal-medial → **ectopic**
- **Male ectopic sites:** bladder neck, verumontanum, **vas/Wolffian derivatives**
- **Female ectopic sites:** **Gartner duct cyst**, vaginal vestibule

CLASSIC TRAPS

- **Trap:** glanular urethra ectodermal — **Actually:** endodermal UGS throughout
- **Trap:** urethral groove opens by apoptosis — **Actually:** proliferation; no apoptosis
- **Trap:** canalisation needs androgen — **Actually:** only fold fusion does
- **Trap:** MIS from Leydig cells — **Actually:** Sertoli; Leydig make testosterone
- **Trap:** gubernacular swelling androgen-dependent — **Actually:** independent; CSL regression and inguinoscrotal phase dependent
- **Trap:** ascending testis congenital — **Actually:** acquired; processus remnant fails to involute
- **Trap:** remove every streak gonad — **Actually:** remove in 46,XY (gonadoblastoma)
- **Trap:** upper vaginal epithelium Müllerian — **Actually:** replaced by UGE to cervix
- **Trap:** OHVIRA = renal agenesis — **Actually:** usually dysplasia; "RA" = renal anomalies
- **Trap:** lower-pole orifice ectopic — **Actually:** lateral, refluxes; upper pole ectopic